Patrick Carpanedo

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Boston, MA

Education

2021 - 2025 **M.S. Computer Science** *Boston University* MA, USA

2016 - 2020 **B.A. in Physics** *College of the Holy Cross* MA, USA

Publications & Presentations

International Conference & Workshop Papers

- Francesco Ciraolo, Mattia Nicolella, **Patrick Carpanedo**, Denis Hoornaert, Marco Caccamo, and Renato Mancuso. 2025. ***Surgical Software-less I/O Virtualization***. https://doi.org/10.1145/3722567.3727847
- Weifan Chen, Ivan Izhbirdeev, Denis Hoornaert, Shahin Roozkhosh, **Patrick Carpanedo**, Sanskriti Sharma, and Renato Mancuso. ***Low-Overhead Online Assessment of Timely Progress as a System Commodity***. https://doi.org/10.4230/LIPIcs.ECRTS.2023.13

Presentation

- Shahin Roozkhosh, Bassel El Mabsout, Cristiano Rodrigues, **Patrick Carpanedo**, Denis Hoornaert, Su Min Tan, Benjamin Lubin, Marco Caccamo, Sandro Pinto, and Renato Mancuso. Burning Fetch Execution: A Framework for Zero-Trust Multi-Party Confidential Computing. In 2024 Technology Innovation Institute (TII) GENZERO Workshop.

Thesis

- **AXI over Ethernet** : This work revolves around using Programmable Logic to export bus-level memory transactions packed into an Ethernet frame to allow methods (e.g. Control Flow Integrity checks, Digital Twinning, and Direct Memory Access) to be processed remotely without kernel intervention.

Work Experience

June 2025 - August 2025 **Embedded Engineer** *Neobotics* Boston, MA, USA

- Collaborated with a cross-functional team to design system architecture for an educational autonomous driving platform through prototyping, research, and strategic planning.
- Conducted comprehensive research on integrated circuits, evaluated technical specifications, developed a central KiCad project, and communicated findings to stakeholders.
- Lead procurement efforts by developing complete parts lists and defining system architecture for a full-scale autonomous driving vehicle, ensuring technical feasibility and integration requirements were met.

January 2022 - August 2025 **Masters Student Researcher** *Cyber Physical Systems Lab* Boston, MA, USA

- Researched and implemented methods for allowing AXI over Ethernet on the Xilinx Zynq Ultrascale+ platform
- Assisted in research to integrate Coresight infrastructure for profiling execution phases in a program
- Assembled and maintained servers (e.g. Ubuntu 24.04 and Proxmox Cluster) to consolidate Xilinx platforms and hardware for facilitated research and collaboration efforts within the lab
- Participated in pseudo-Technical Program Committee (TPC) meetings with Lead P.I. to review papers for high tier systems conferences
- Volunteered to assist or lead students enrolled in directed studies inside of CPS lab.

May 2019 - July 2019 **Research Assistant** *College of the Holy Cross* Worcester, MA, USA

- Assembled and verified electrical tolerances and timings for subsystems of a custom Beam Profile Monitor (BPM). Evaluated the BPM system through experiments with LabView and oscilloscopes, logged findings, and facilitated weekly presentations and discussions with other research groups.

Teaching and Mentoring

January 2024 - August 2025 **F1Tenth Study Mentor** *Boston University* Boston, MA, USA

- Assisted undergraduates with F1Tenth hardware integration on Jetson Nano with ROS2, taught electronic design basics, and ensured safe handling of high-current and sensitive electronics.

August 2023 - December 2023 **UR2PhD Mentor** *Computing Research Association* Boston University

- Developed mentoring skills, led group and individual sessions with undergraduates to create PoV Display hardware/software modules for the KV260, sourced and verified components, and trained students in academic research methods.

Skills

- **Languages: English** [Native], **Portuguese** [Fluent], **Spanish** [Fluent]
- **Programming:** C, C++, Java, Python, Golang, SpinalHDL, SQL, Scala
- **Design:** SpinalHDL, Verilog, CAD, PCB design, Carpentry, Fabrication
- **Hardware Debugging:** Xilinx Integrated Logic Analyzer, ARM Coresight, Circuit Debugging
- **System Administration:** Network Architecture, Proxmox, Incus, Kubernetes, Docker, Git, Cluster Management